```
1   PROGRAM PLC_PRG
2   VAR
3       // Panel (inputs)
4       Start : BOOL ;
5       Stop : BOOL ;
6       Reset : BOOL ;
7       E_Stop : BOOL ;
8
9       // Sensors (inputs)
10      High_Sensor : BOOL ;
11      Low_Sensor : BOOL ;
12      Pallet_Sensor : BOOL ;
13      Loaded : BOOL ;
14      At_Entry_Left : BOOL ;
15      At_Entry_Right : BOOL ;
16      At_left_Exit : BOOL ;
17      At_right_Exit : BOOL ;
18
19      // Actuators (outputs)
20      Conveyor_Entry : BOOL ;
21      Load_Conveyor : BOOL ;
22      Transfer_Left : BOOL ;
23      Transfer_Right : BOOL ;
24      Conveyor_Left : BOOL ;
25      Conveyor_Right : BOOL ;
26
27      // Display (output)
28      Display_Dig : WORD ;
29
30      // System state
31      System_Running : BOOL ;
32      E_StopLatched : BOOL ;
33      E_StopUnLatch : BOOL ;
34      E_StopL1 : BOOL ;
35      E_StopL2 : BOOL ;
```

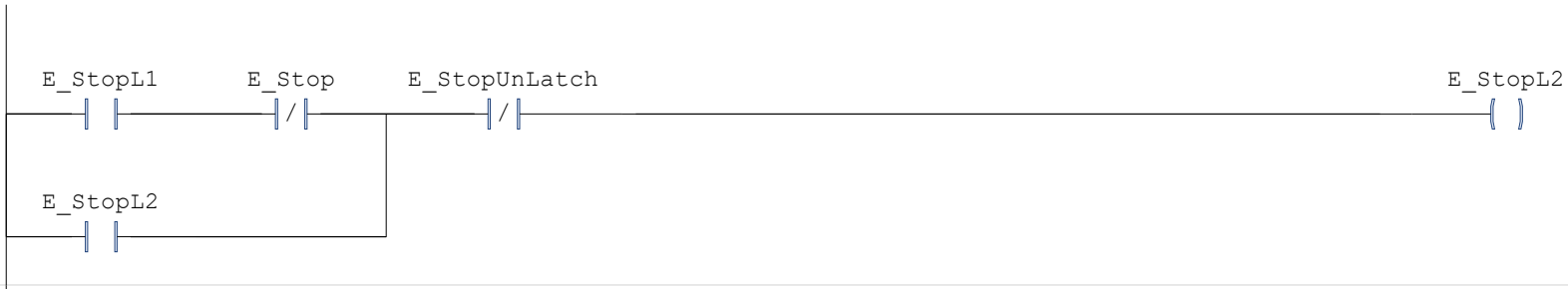
```
36
37          // Process state
38          Entry_Enable :  BOOL ;
39          Loading_Active :  BOOL ;
40          Stop_Conv_Entry :  BOOL ;
41          High_Item :  BOOL ;
42          Stop_Transfer_L :  BOOL ;
43          Stop_Transfer_R :  BOOL ;
44
45          // Function block instances
46          Height_Sorter_HIGH :  Height_Sorter ;
47          Height_Sorter_Low :  Height_Sorter ;
48          Side_Conveyor_Left :  Side_Conveyor ;
49          Side_Conveyor_Right :  Side_Conveyor ;
50          Sorting_Counter_0 :  Sorting_Counter ;
51      END_VAR
52
```

1

E-Stop: signal confirmation and latch



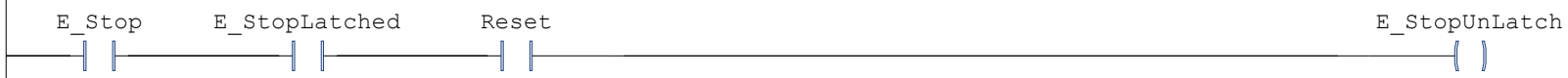
2



3



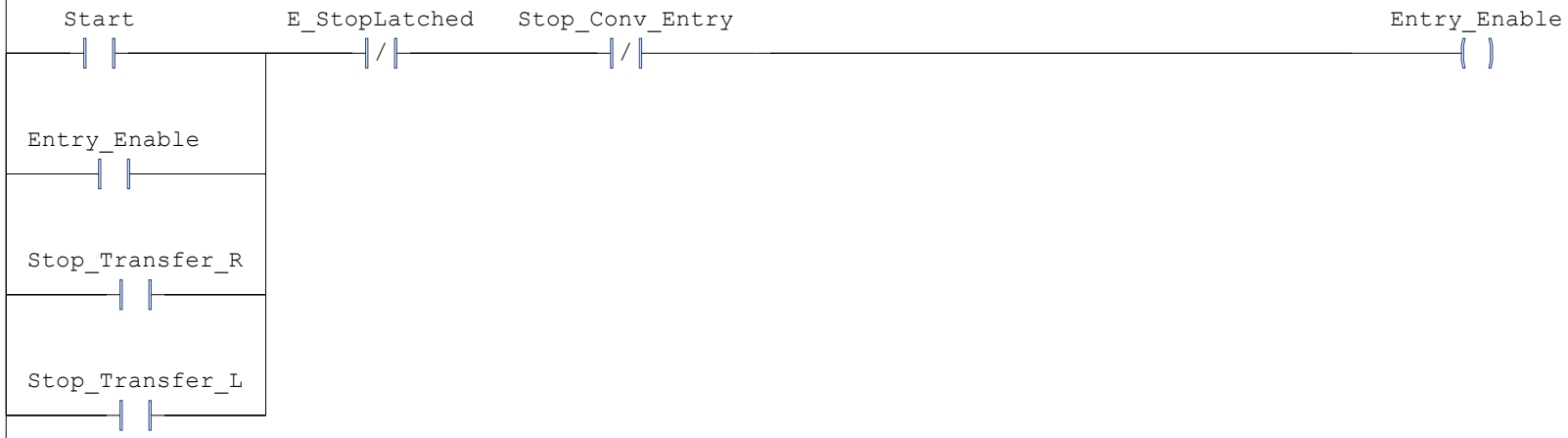
4 *Reset releases E-Stop latch*



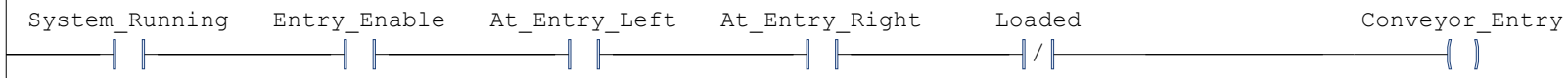
5 *System start/stop (System_Running)*



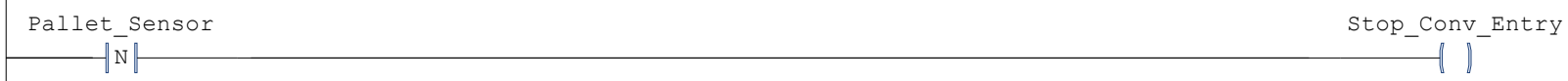
6 *Entry conveyor enable*



7 *Entry conveyor output*



8 *Pallet detected: stop entry conveyor*



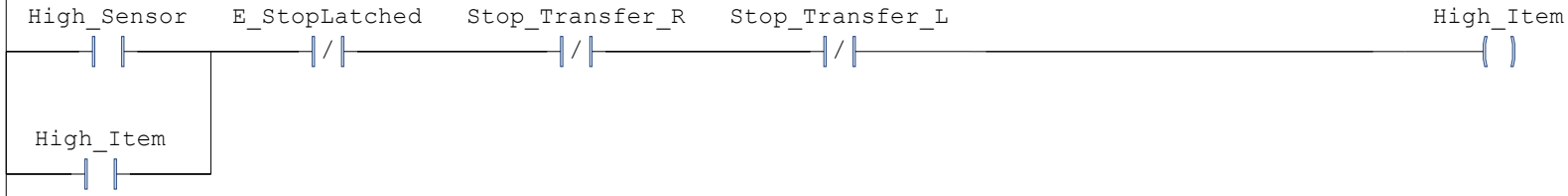
9 *Loading sequence latch*



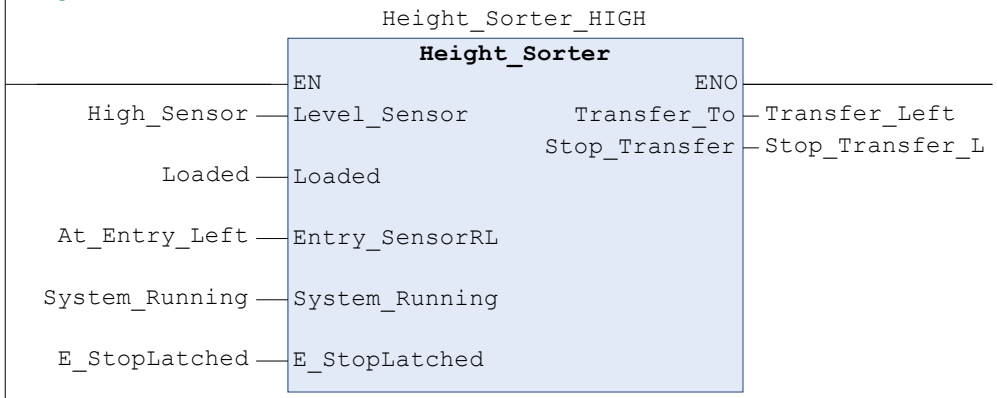
10 *Load conveyor output*



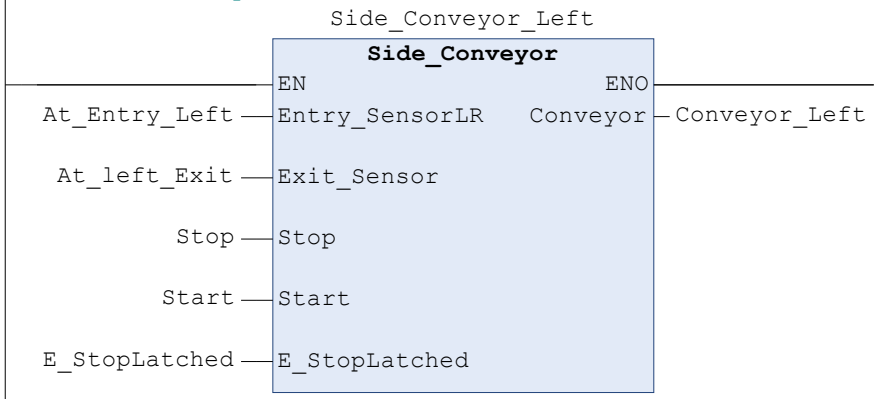
11 *High item latch (High_Item)*



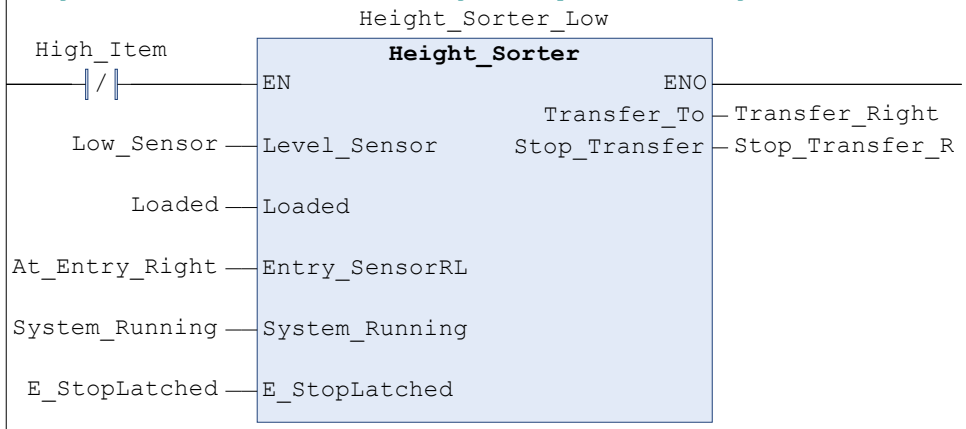
12 *Height sorter HIGH: transfer left*



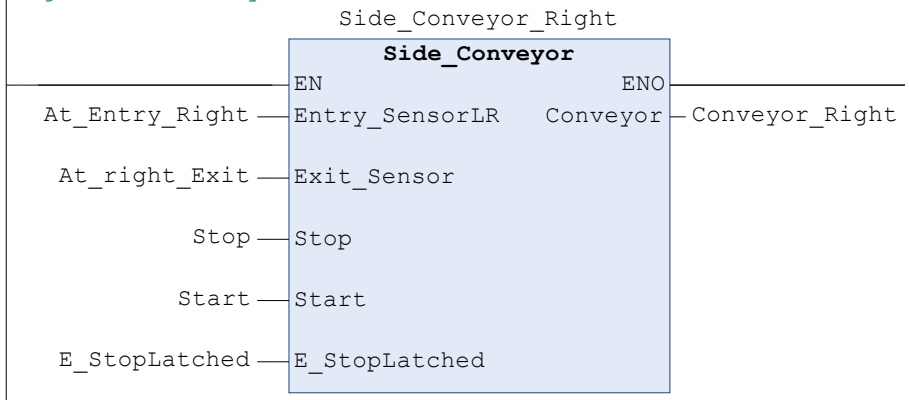
13 *Left side conveyor*



14 *Height sorter LOW: transfer right (depends on rung 11)*



15 *Right side conveyor*



16

Sorted item counter and display

